

Formation of dioxins during exposure of pesticide formulations to sunlight.



Chlorinated pesticides can contain impurities of dibenzo-p-dioxins and dibenzofurans (PCDD/Fs), and their precursors, as a result of various manufacturing processes and conditions. As precursor formation of PCDD/Fs can also be mediated by ultraviolet light (UV), this study investigated whether PCDD/Fs are formed when currently used pesticides are exposed to natural sunlight. Formulations containing pentachloronitrobenzene (PCNB; n=2) and 2,4-dichlorophenoxyacetic acid (2,4-D; n=1) were exposed to sunlight in quartz tubes, and the concentration of 93 PCDD/F congeners were monitored over time. Considerable formation of PCDD/Fs was observed in both PCNB formulations (by up to 5600%, to a maximum concentration of 57000 $\mu\text{g } \Sigma\text{PCDD/F kg}(-1)$) as well as the 2,4-D formulation (by 3000%, to 140 $\mu\text{g } \Sigma\text{PCDD/F kg}(-1)$). TEQ also increased by up to 980%, to a maximum concentration of 28 $\mu\text{g kg}(-1)$ in PCNB, but did not change in the 2,4-D formulation. Assuming similar yields as observed in the present study as a worst case scenario the use of PCNB in Australia may result in the formation of 155 g TEQ annum⁻¹, contributed primarily by OCDD formation. This warrants detailed evaluations on the contemporary release of PCDD/Fs to the environment after the use of pesticides. Changes in congener profiles (including the ratio of PCDDs to PCDFs (DF ratio)) suggest that pesticide sources of PCDD/Fs after sunlight exposure may not be recognized based on matching source fingerprints established from manufacturing impurities. These changes also provide preliminary insights into the possible formation routes and types of precursors involved. Copyright © 2012 Elsevier Ltd. All rights reserved.